DigiCow Farmer Adoption Prediction

Predicting the probability of agricultural practice adoption

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Zindi Score Achieved

0.9538

Composite score: $0.75 \times (1 - LogLoss) + 0.25 \times AUC$

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1 Project Overview

1.1 Context and Objective

This Zindi competition, organised by **DigiCow**, aims to predict the probability that a farmer will adopt a practice after their first training session. DigiCow is a digital agricultural extension platform seeking to maximise the impact of its training programmes by targeting farmers most likely to adopt new practices.

Concrete objectives for DigiCow:

- Prioritise follow-up support for farmers with a high adoption probability
- Tailor extension services based on predicted farmer profiles
- Improve training programmes to increase real-world impact

1.2 Prediction Task

The model must produce, for each farmer, **three adoption probabilities**:

- $P(\text{adoption} \leq 7 \text{ days})$
- $P(\text{adoption} \leq 90 \text{ days})$
- $P(\text{adoption} \leq 120 \text{ days})$

Key constraint: only information available *at the time* of the training session may be used — any future data leakage is strictly forbidden.

1.3 Evaluation Metric

The official metric is a composite score defined as:

$$\text{Score} = 0.75 \times (1 - \text{LogLoss}) + 0.25 \times \text{AUC-ROC} \tag{1}$$

This formulation rewards models that produce well-calibrated probability estimates (via LogLoss) while also valuing discriminative power (AUC).

1.4 Datasets

Table 1: Dataset description

File	Rows	Columns	Description
`Train.csv`	13,536	17	Training data with target labels
`Test.csv`	5,621	14	Test data (no targets)
`Prior.csv`	44,882	17	Historical session data
`SampleSubmission.csv`	5,621	7	Submission format

Available columns include: farmer ID, training date, gender, age, registration method, cooperative membership, county, subcounty, ward, group, trainer, topics list, and the three binary targets.

2 Exploratory Data Analysis (EDA)

2.1 Class Imbalance

The problem exhibits a strong class imbalance, typical of behavioural adoption tasks:

Table 2: Adoption rates in the training set

Target	Positives	Total	Rate
Adoption within 7 days	153	13,536	1.13%
Adoption within 90 days	214	13,536	1.58%
Adoption within 120 days	302	13,536	2.23%

This extreme imbalance (fewer than 2% positives for short horizons) justifies the use of `class_weight` and `scale_pos_weight` in the models, and the prioritisation of LogLoss as the primary metric.

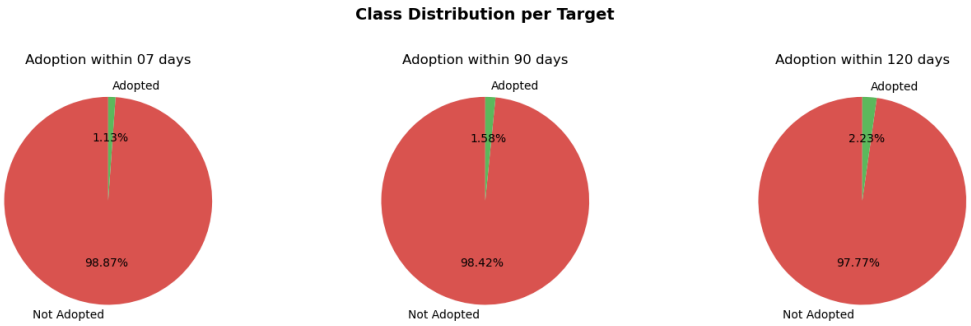


Figure 1: Class distribution for the three time horizons

2.2 Demographic Features

Analysis of categorical variables reveals the composition of the farmer base:

- **Gender**: 9,541 female (70.5%) vs 3,995 male (29.5%)
- **Age**: 10,828 farmers above 35 years old (80%) vs 2,708 below 35
- **Registration**: 8,383 via USSD (62%) vs 5,153 manual
- **Cooperative**: 1,928 members (14.2%) vs 11,608 non-members

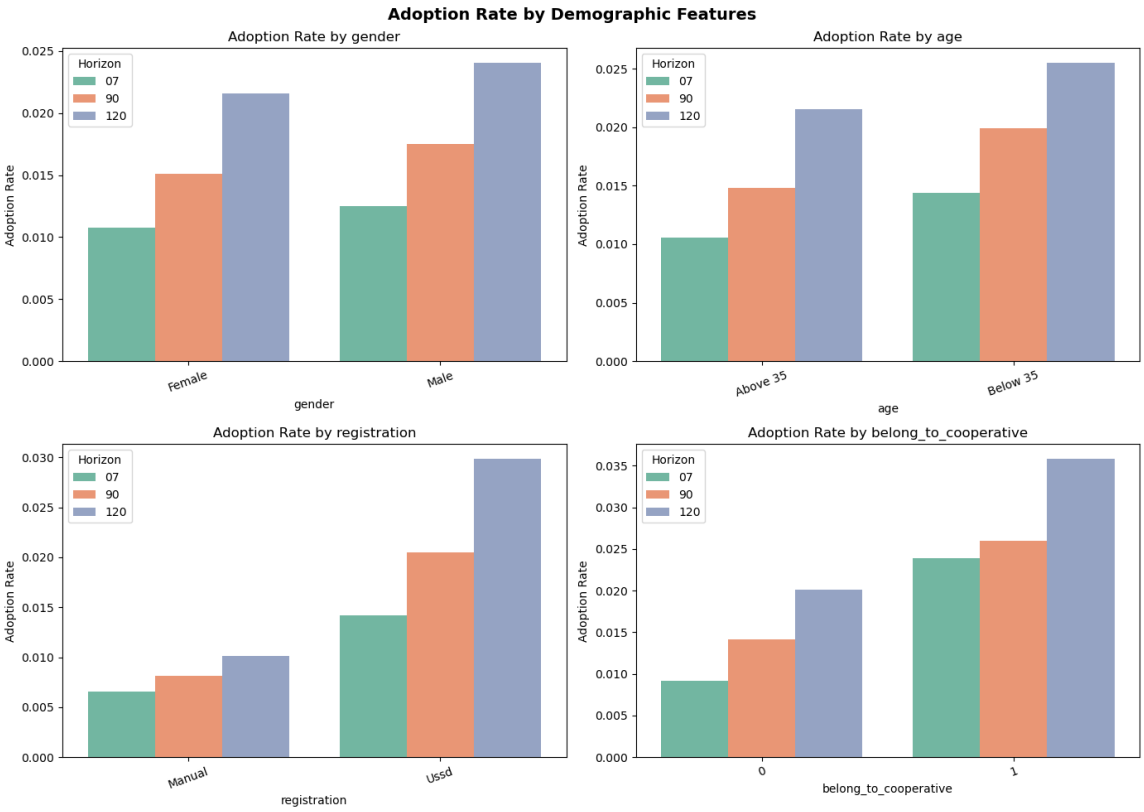


Figure 2: Adoption rates by demographic features across the three time horizons

2.3 Geographic Distribution

Table 3: Geographic statistics

Variable	Unique values
Counties	8
Subcounties	26
Wards	66
Groups	864
Farmers	13,536

Counties show significant adoption disparities. County `CNT_mvqqmw` (734 observations) has the highest rates (3.8% at 7 days), while others such as `CNT_vvuukv` (3,049 observations) remain below 0.5%.

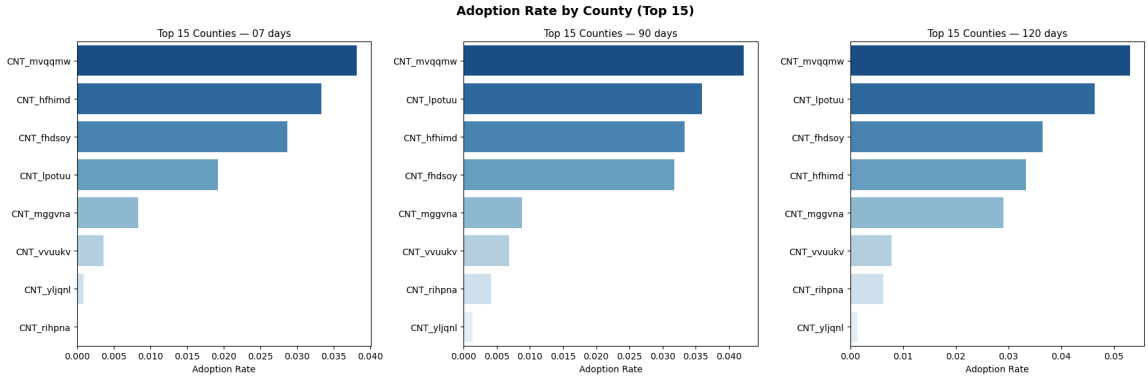


Figure 3: Top 15 counties by adoption rate across the three time horizons

2.4 Training Topics Analysis

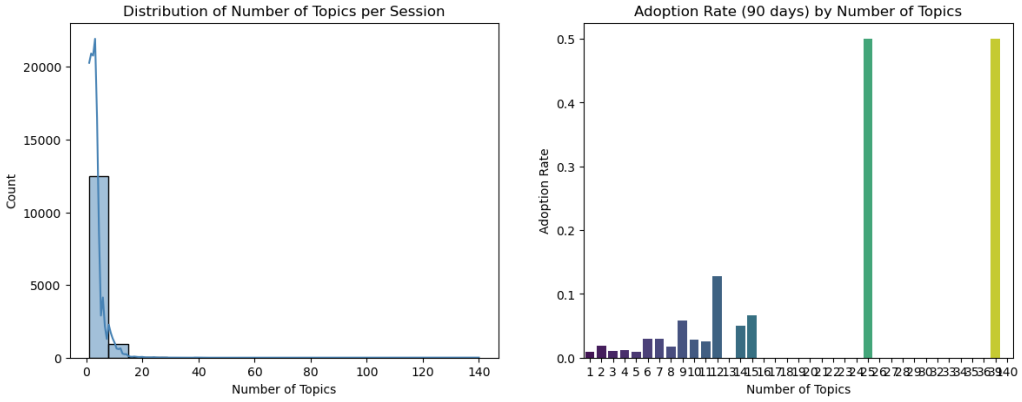


Figure 4: Distribution of number of topics per session (left) and adoption rate (90 days) by number of topics (right)

The figure shows that a moderate number of topics per session correlates with higher adoption rates, suggesting that focused sessions (2–3 topics) are more effective than overloaded ones.

2.5 Numerical Feature Distributions

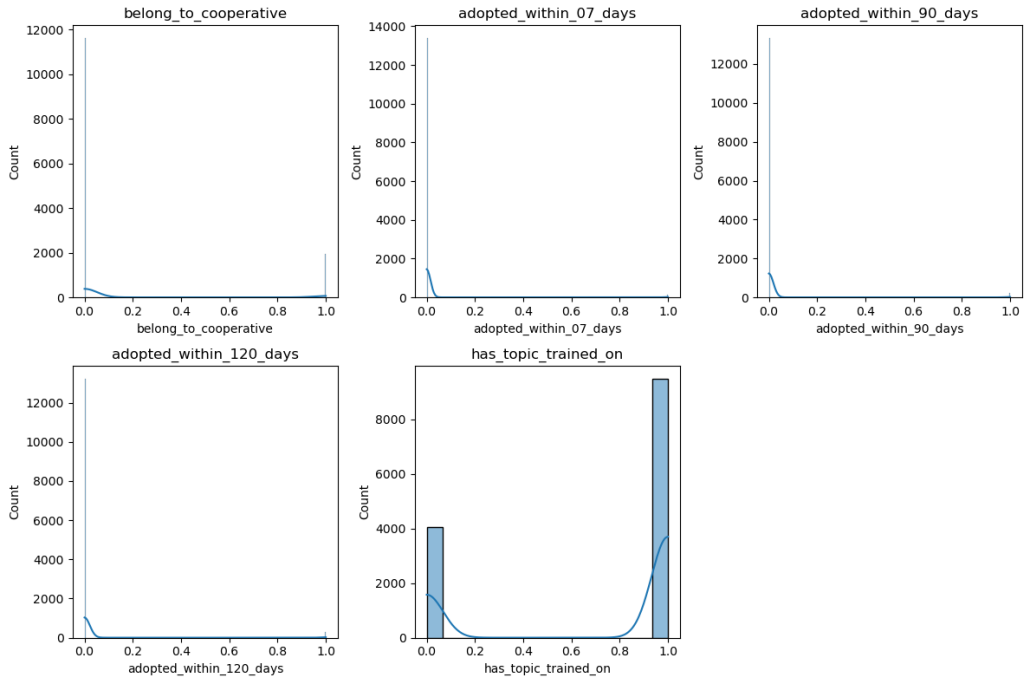


Figure 5: Distribution of main numerical variables in the training set

3 Feature Engineering

3.1 Overview

Feature engineering is the cornerstone of this solution. A total of **293 features** were constructed, organised into several functional groups.

3.2 Historical Statistics (Prior Stats)

The `Prior.csv` file (44,882 historical sessions) is used to build **smoothed target encodings**. For each categorical variable c and each target t, the smoothed rate is computed as:

$$\hat{r}_{c,t} = \frac{\sum_{i \in c} y_{i,t} + K \cdot \bar{y}_t}{n_c + K} \tag{2}$$

where $K = 50$ is the smoothing strength, $\bar{y}_t$ is the global adoption rate for target t, and n_c is the number of observations in category c. This shrinkage toward the global mean prevents overfitting on rare categories.

Statistics are computed for: `county`, `subcounty`, `ward`, `group_name`, `trainer`, `gender`, `age`, `registration`, `month`, `dayofweek`, `quarter`, `belong_to_cooperative`, `has_topic_trained_on`, `year_month`, `week`.

3.3 Out-of-Fold (OOF) Encoding

To prevent data leakage, high-cardinality categorical variables (`group_name`, `county`, etc.) are encoded using an **Out-of-Fold** (OOF) scheme with 5-fold stratified cross-validation:

3.4 Feature Groups

Table 4: Feature groups and descriptions

Group	Description
Temporal	Cyclic sin/cos encoding for month and day of week, normalised time position, weekend flag, quarter
Demographic	Numerical encoding of age, gender, registration method, cooperative membership
Topics	Topic count, Shannon entropy, category flags, smoothed rates per topic
Trainer	Trainer quality score (smoothed adoption rate), consistency, event count
Geographic	Smoothed and OOF rates for county, ward, sub-county
Farmer history	Number of prior sessions, recency, historical adoption rate
Group	Group size, cohesion score
Interactions	Cross-feature products, cross-target ratios (07/90, 07/120, 90/120)

3.5 Shannon Entropy of Topics

An original feature measures the **thematic diversity** of each session via Shannon entropy:

$$H(\text{topics}) = -\sum_{i} p_i \log p_i \tag{3}$$

Zero entropy indicates a single topic (focused session), while high entropy signals a broad mix of subjects.

3.6 Feature Correlation with Targets

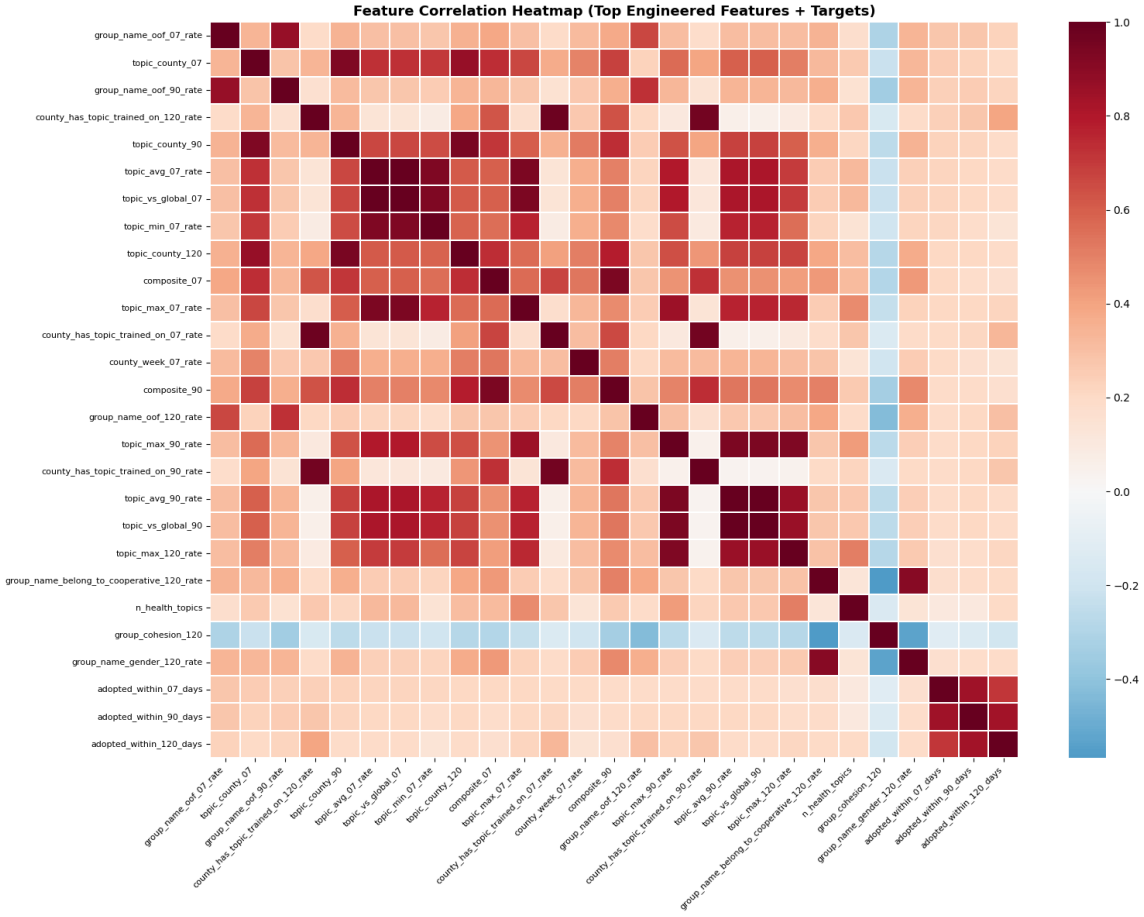


Figure 6: Correlation heatmap of top engineered features with the three targets

The features most correlated with the 7-day target are:

1. `group_name_oof_07_rate` (0.284)
2. `topic_county_07` (0.259)
3. `group_name_oof_90_rate` (0.239)
4. `county_has_topic_trained_on_120_rate` (0.236)
5. `topic_county_90` (0.234)

3.7 Train vs Test Distribution

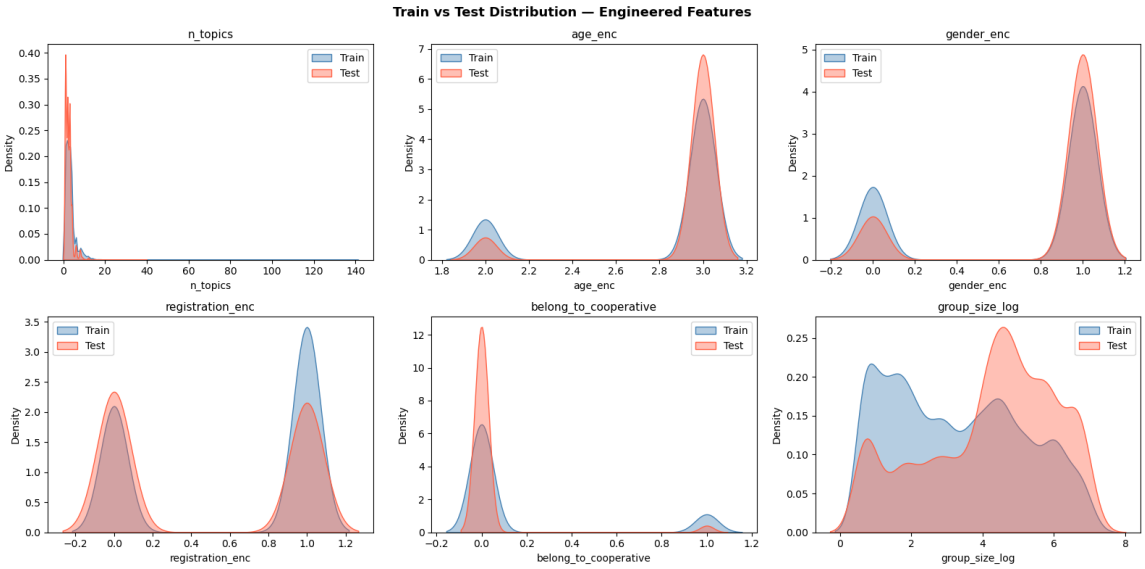


Figure 7: Distribution of key engineered features in train and test sets — no major data shift observed

The figure confirms that train and test distributions are consistent, validating the robustness of the feature engineering pipeline.

4 Modelling

4.1 Overall Architecture

The solution is based on a **three-level stacking** architecture:

1. **Level 1**: Ensemble of 8 diverse models trained with cross-validation
2. **Pseudo-labeling**: Training set augmentation using high-confidence test predictions
3. **Level 2 (Meta-learner)**: Stacking with XGBoost and Logistic Regression
4. **Post-processing**: Monotonicity constraint and probability clipping

4.2 Level 1 Models

Table 5: Models used in the level-1 ensemble

Model	Key configuration
`hgb1` (HistGradientBoosting)	`max_iter=800, lr=0.03, max_depth=6, class_weight=balanced`
`hgb2` (HistGradientBoosting)	`max_iter=800, lr=0.05, max_depth=8, class_weight=balanced`
`rf` (Random Forest)	`n_estimators=400, max_depth=12, class_weight={0:1, 1:pw}`
`et` (Extra Trees)	`n_estimators=400, fully random splits`
`lgbm_fixed` (LightGBM)	`n_est=3000, lr=0.02, num_leaves=63, scale_pos_weight=pw`
`lgbm_fixed2` (LightGBM)	`n_est=3000, lr=0.02, num_leaves=31, max_depth=5`
`xgb_fixed` (XGBoost)	Fixed parameters with scale_pos_weight
`catboost` (CatBoost)	Target-specific configuration

Handling class imbalance: The `scale_pos_weight` parameter (pw) is tuned per target: $pw = 80$ for 7 days, $pw = 60$ for 90 days, $pw = 40$ for 120 days.

4.3 Hyperparameter Tuning (Optuna)

LightGBM and XGBoost hyperparameters are optimised with **Optuna** using the TPE (Tree-structured Parzen Estimator) algorithm over 50 trials. The search space includes:

- `n_estimators`: [500, 4000]
- `learning_rate`: $[0.005, 0.1]$ (log-scale)
- `num_leaves`: [15, 127]
- `max_depth`: [3, 9]
- `min_child_samples`: [10, 100]
- Regularisation (α, λ) and subsampling parameters

Best parameters found for the 7-day target:

- LightGBM: `n_est=3170, lr=0.067, num_leaves=72, max_depth=7`
- XGBoost: `n_est=3981, lr=0.069, max_depth=9`

4.4 Stratified Cross-Validation

Training uses **5-fold stratified cross-validation** (`StratifiedKFold`, `random_state=42`). Stratification ensures that each fold preserves the positive class proportion, which is critical given the severe class imbalance.

4.5 Pseudo-Labeling

Pseudo-labeling augments the training set with high-confidence test examples:

- **Pseudo-positive labels**: $\hat{p} \geq$ PSEUDO_THR_HI (high threshold)
- **Pseudo-negative labels**: $\hat{p} \leq$ PSEUDO_THR_LO (low threshold)

Final predictions blend initial and pseudo-labelled predictions at a 70/30 ratio.

4.6 Stacking (Meta-learning)

OOF predictions from all 8 level-1 models are used as features to train two meta-learners:

- **Meta-XGBoost**: `max_depth=2, learning_rate=0.03, n_estimators=300`
- **Meta-Logistic Regression**: `C=0.05`

The final stacking prediction is a 50/50 weighted average of the two meta-learners.

5 Post-processing

5.1 Monotonicity Constraint

By causal logic, the probability of adoption cannot decrease over time:

$$P(\text{adoption} \leq 7) \leq P(\text{adoption} \leq 90) \leq P(\text{adoption} \leq 120) \tag{4}$$

This constraint is enforced via a cumulative maximum:

```
def enforce_monotonicity(pred_07, pred_90, pred_120):
    pred_90  = np.maximum(pred_90,  pred_07)
    pred_120 = np.maximum(pred_120, pred_90)
    return pred_07, pred_90, pred_120
```

Listing 1: Enforcing monotonicity

5.2 Prediction Clipping

To prevent infinite LogLoss from predictions at exactly 0 or 1, all probabilities are clipped to $[0.001, 0.999]$:

$$\hat{p} \leftarrow \text{clip}(\hat{p}, \epsilon, 1-\epsilon), \quad \epsilon = 0.001 \tag{5}$$

6 Results

6.1 OOF (Out-of-Fold) Performance

Table 6: Final results per target — training set (OOF)

Target	LogLoss	AUC-ROC	F1 (best)	Estimated Score
Adoption 7 days	0.0180	0.9929	0.6757	0.9847
Adoption 90 days	0.0239	0.9880	0.7368	0.9790
Adoption 120 days	0.0280	0.9881	0.7707	0.9760
Average	**0.0233**	**0.9897**	**0.7277**	**0.9799**

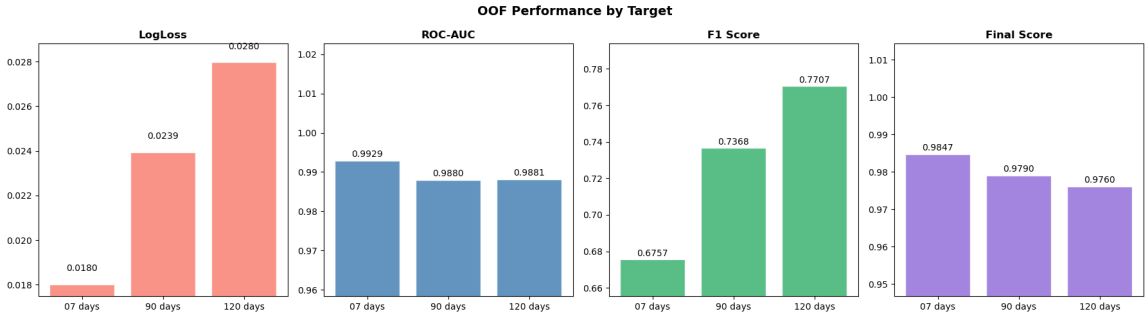


Figure 8: OOF metrics per target: LogLoss, AUC-ROC, F1-Score and composite score

6.2 Zindi Score

After submission on the Zindi platform, the score obtained on the test set is:

Zindi Score (public test)

0.9538

Estimated OOF score: 0.9799

The gap between the OOF score (0.9799) and the Zindi score (0.9538) is attributable to mild overfitting to the training set, which is common in severely imbalanced problems where OOF scores tend to be optimistic.

6.3 Submission Sample

Sample predictions from the first rows of the submission file:

Table 7: Submission file excerpt

ID	P(7 days)	P(90 days)	P(120 days)
ID_LEG1GM	0.00100	0.00100	0.00100
ID_1UKOKW	0.00100	0.00100	0.00100
ID_U5H2YK	0.04698	0.04698	0.04698
ID_55957A	0.04698	0.04698	0.04698
ID_N1AC0A	0.00100	0.00100	0.00100

7 Discussion and Conclusion

7.1 Strengths of the Solution

- **Rich feature engineering**: 293 features leveraging historical data (`Prior.csv`), leak-free OOF encoding, cyclic temporal features and cross-feature interactions.
- **Diverse ensemble**: 8 models from different families (gradient boosting, random forests, extreme trees) capturing varied patterns.
- **Robustness to imbalance**: Target-specific `scale_pos_weight`, stratified encoding throughout.
- **Temporally coherent post-processing**: The monotonicity constraint ensures logically consistent probability estimates across time horizons.

7.2 Potential Improvements

- Reduce OOF-test overfitting via stronger regularisation or more cross-validation folds
- Explore survival analysis models better suited to the temporal nature of the targets
- Incorporate embeddings for text-like variables (topic names, group names)
- Deepen probability calibration (Platt scaling, isotonic regression)

7.3 Conclusion

This solution achieves a competitive score of **0.9538** on Zindi through a combination of in-depth feature engineering exploiting historical data, an ensemble of 8 heterogeneous models optimised with Optuna, and a two-level stacking approach. Careful handling of the extreme class imbalance and temporal coherence constraints contribute significantly to the quality of the predictions.